Al-Balqa Applied University

جامعة البلقاء التطبيقية

Midterm Exam

Course Title	General Chemistry lab		Course Code	30206111	
Semester	1st		Academic Year	2023-2024	
Exam Date	22/11/2023	Day	Wednesday	Time	2:30-3:20

الخميس 8:30 - 11:30

Student Nam[redacted] Student I[redacted] ction

اسم المدرس : لينا

GENERAL INSTRUCTIONS

1. Students are not allowed to leave the exam room during the first 30 minutes of the exam.
2. Late students are not permitted to complete their exams after the first 30 minutes unless authorized by the proctor (no students left the exam session), knowing that no additional time will be granted.
3. Students must use a blue or black pen. Pencils are not allowed.
4. Consumption of any food or beverage is prohibited except in medical cases.
5. All electronic devices (e.g., cell phones) are prohibited.
6. All cell phones must be turned off and placed on the invigilator's desk in the exam room.
7. All notes and reading materials must be placed on the invigilator's desk in the exam room.
8. Students are not permitted to speak to or ask questions of their classmates. If a student has a question, he or she must raise his or her hand to be assisted by the proctor.
9. Write the full information required on this page before commencing your exam.
10. Students are not allowed to leave the room once the exam commences for any reason unless it is absolutely necessary.
11. Reported student cheating cases, misbehavior, or other student misconduct on examination sessions will be reported to the Investigations Committee as per internal bylaws.

I have understood and abide with the exam policy. Student's signature:

Question No.	Question Type	CLOs assessed	Marks Weight	Scored Marks	Marker signature	Student Score
1-15	Multiple Choice	CLIO1-5	30(100%)			26 V.Good
Total			30(100%)			30
Moderation Review						
Comments						
Marks Reviewed by				Signature & Date		

Use the following data when needed R=0.08206L.atm/K.mol , 1atm=760torr=760mmHg .

Atomic masses (g/mol) = H:1 ,O:16 ,C:12 ,N:14 , Cl:35.5 ,S:32 ,Ca:40

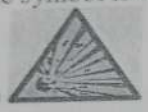
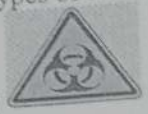
Choose the correct answer of the following question's and transfer it in CAPITAL letter to table

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	C	B	D	A	B	A	C	C	C	B	A	C	A	D	D

Q1- Which tool is used to measure the volume of a liquid?

A) Pipette B) Microscope C) Graduated cylinder D) Balance

Q2- The symbol is used to describe flammable material types of substances?

A)  B)  C)  D) 

Q3- What is the volume of the liquid in this graduated cylinder?



A) 20.0 mL B) 29.5 mL C) 44.5 mL D) 21.5 mL

Q4- The density of gold is 19.3 g/mL. Three identical cubes of gold are inserted in a graduated cylinder containing water. The initial reading of water is 15.0 mL, and the final reading is 20.0 mL. Calculate mass in grams per one cube of gold.

A) 32.1 B) 24.1 C) 19.3 D)16.0

Q5- Based on what you studied about solubility rules, which of the following is a correct statement ?

A) Mixing I_2 with Chloroform or hexane ends up with I_2 precipitated in the bottom of test tube

B) Chloroform and hexane with water do not mix and hence form two separate layers.

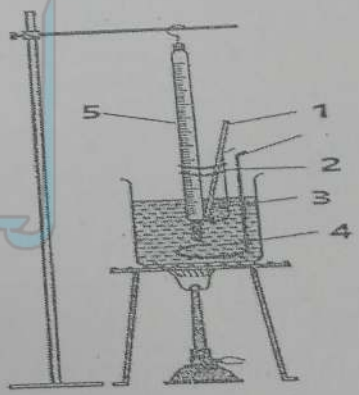
C) Glucose is insoluble in chloroform because glucose has a long nonpolar hydrocarbon chain.

D) Ammonia (NH_3) dissolves in cyclohexanol ($C_6H_{11}OH$) because both are polar.

Q6- $CaCO_3$ salt is ---------- in water (H_2O).

A) Soluble B) Insoluble

Q7- In the next figure what is the number and name of the tool that used to used measured temperature of liquid?



A) 2, Thermometer

B) 3, Test tube

C) 5, Thermometer

D) 1, Capillary tube

Q8- At which of the following external pressures will benzene has its lowest boiling point?

A) 2.5 atm B) 1.95 atm C) 0.92 atm D) 1 atm

Q9- Which of the following statement is incorrect?

A) $CH_3CH_2CH_2OH$ lower boiling point than $CH_3CH_2CH_2CH_2CH_2OH$
B) $CH_3CH_2CH_2OH$ higher boiling point than $CH_3CH_2CH_3$
C) CH_3OH lower boiling point than CH_3CH_3
D) $CH_3CH_2CH_2OH$ higher boiling point than $CH_3CH_2CH_2CH_3$

Q10- A hydrated salt of the type $CaCl_2.XH_2O$ has 32.5% water by mass. Its formula is……

A) $CaCl_2.H_2O$ B)$CaCl_2.3H_2O$ C)$CaCl_2.4H_2O$ D) $CaCl_2.2H_2O$

Q11- The equipment that was used in water of crystallization experiment to cool the sample in is:

A) Desiccator B) Crucible C)Tung D)Balance

Q12- The data in the table was collected during water of crystallization experiment, use it to calculate the percentage weight of water in an Alum.

A) $H_2O\% = 37\%$
B) $H_2O\% = 42\%$
C) $H_2O\% = 52\%$
D) $H_2O\% = 46\%$

Mass of crucible + cover+ Alum	**40.23 g**
Mass of crucible + cover	**39.23 g**
Mass of crucible + cover+ Alum after heating	**39.81 g**

Q13- A sample of an unknown volatile liquid was heated using Dumas method use the data in the table to decide which one the unknown volatile liquid was

A) C_6H_{12}
B) C_6H_{14}
C) C_7H_{14}
D) C_7H_{16}

Mass of empty flask , aluminum foil and rubber band	27.0928 g
Mass of flask , aluminum foil , rubber band and the vapour	27.4593 g
The atmospheric pressure	1.001 atm
temperature	18.0 °C,
Volume of the flask	0.1040 L

Q14- Which of the following cases causes an increase in the calculated molar mass of the volatile liquid?

A) The reported temperature is 5 °C lower than the actual temperature.
B) The reported pressure is higher than the actual one.
C) If the value of R is considered lower than the true value
D) The measured volume is lower than the actual one by 20 mL.

$$M.m = \frac{mRT}{VP}$$

Q15- The density of NO_2 in a 4.50 L tank at 760.0 torr and 25.0°C is _____ g/L.

A) 1.64
B) 1.30
C) 1.68
D) 1.88

GOOD LUCK